



**DEPARTMENT OF THE AIR FORCE
HEADQUARTERS 147TH ATTACK WING
HOUSTON TX 77034**

MEMORANDUM FOR CONTRACTING

FROM: 147 ATKW/CS
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SUBJECT: Sole Source Justification for the Purchase of Rohde & Schwarz PRCEP3_HE600 Spectrum Analyzer

1. **PURPOSE:** This memorandum provides the justification for the sole source procurement of one (1) PRCEP3_HE600 Signal and Spectrum Analyzer from the manufacturer, Rohde & Schwarz. This equipment is critical for mission-essential testing and analysis of advanced electronic warfare (EW) and radar systems.
2. **BACKGROUND:** Our unit is tasked with the verification and validation of next-generation airborne surveillance and defense systems. This requires precise measurement of complex, low-power, and frequency-agile signals in a dense electromagnetic environment. The technical specifications of the signals under test demand equipment with capabilities that exceed those of standard, general-purpose spectrum analyzers currently available in our inventory or through other vendors.
3. **SOLE SOURCE JUSTIFICATION:** Extensive market research, including research at Keysight Technologies, Anritsu, and Tektronix, has determined that only the Rohde & Schwarz PRCEP3_HE600 meets the government's minimum technical requirements. The unique features of this specific analyzer are not available from any other source. The salient characteristics are as follows:
 - a. **Unmatched Phase Noise Performance:** The PRCEP3_HE600 offers an exceptionally low phase noise floor, which is a critical discriminator for accurately characterizing the performance of advanced low-probability-of-intercept (LPI) radar and communication systems. Competing models do not meet our required noise threshold for this sensitive task.
 - b. **Real-time Analysis Bandwidth:** This analyzer provides a wide, real-time analysis bandwidth that is essential for the seamless capture and analysis of elusive, transient signals and frequency-hopping patterns used in modern EW systems. This capability is proprietary to Rohde & Schwarz and is not matched by competitors.
 - c. **Proprietary Measurement Applications:** The required analysis is dependent on specific firmware and software applications (e.g., R&S@FSW-K6 Pulse Measurement) that are exclusive to the Rohde & Schwarz platform. These applications automate complex measurements and are essential for meeting our testing timelines and accuracy requirements. No other vendor provides equivalent, integrated tools.

4. **IMPACT OF NON-APPROVAL:** Failure to procure the Rohde & Schwarz PRCEP3_HE600 will result in a significant mission capability gap. Without it, we will be unable to validate the performance of critical defense systems against emerging threats, potentially leading to unacceptable operational risks and delays in system deployment. Utilizing alternative, less-capable equipment would yield inaccurate data and fail to meet the project's minimum testing requirements.
5. **CONCLUSION:** Due to the unique combination of superior phase noise performance, real-time analysis bandwidth, and proprietary software, Rohde & Schwarz is the only vendor capable of meeting our mission requirements. Therefore, it is in the best interest of the U.S. Air Force to approve this sole source procurement.

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